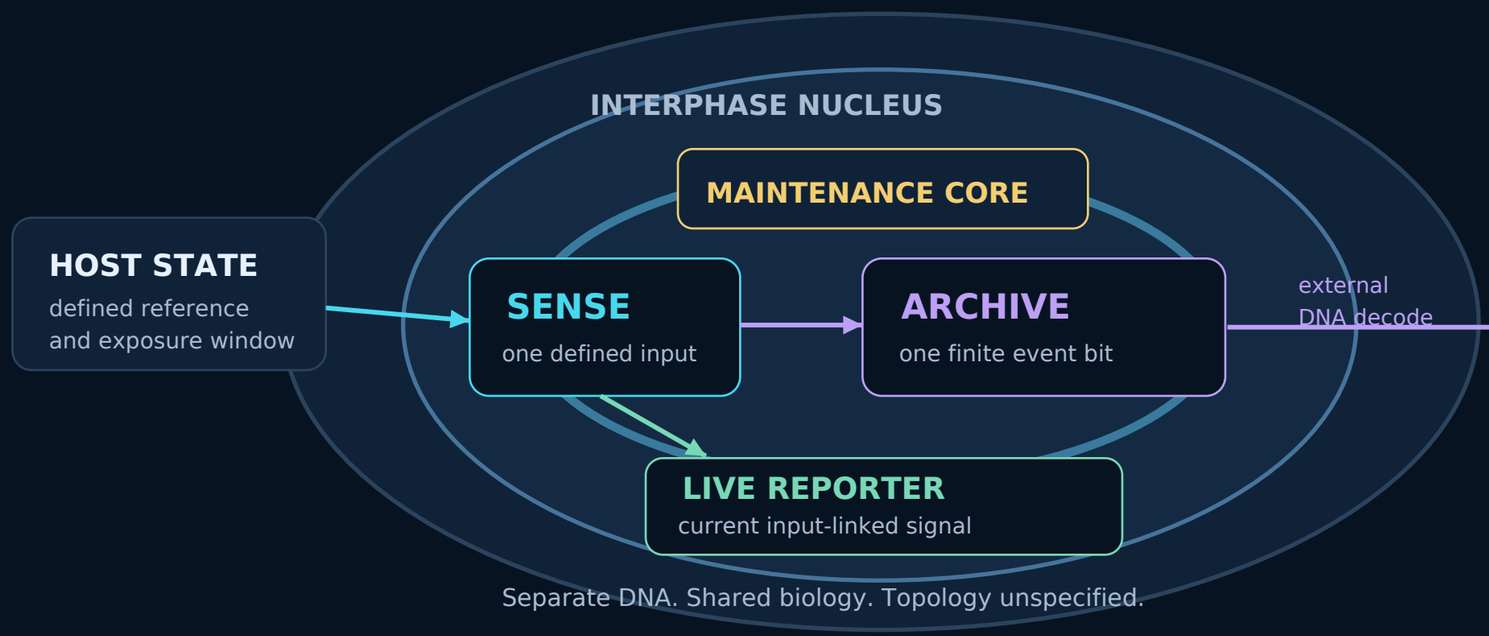
A cellular history recorder

COMPONENT PRECEDENTS / SYSTEM UNVALIDATED

Live state. Finite DNA memory. Evidence attached to every interpretation.

HOST SUPPLIES ENERGY, EXPRESSION, DNA REPLICATION AND REPAIR



OUTSIDE THE CELL

Evidence ledger

Physical location + integrity
Functional competence + memory
Host impact + missing outcomes

INTERPRETATION

RECORDED

QUALIFIED BLANK

UNKNOWN

Assay unit, window and uncertainty are explicit.

01 FINITE HISTORY

One event latch supports “at least one qualifying event.” Counts and timestamps require further development.

02 DISTINCT ENDPOINTS

Output OFF, residual signal, physical DNA clearance and host recovery require separate evidence.

03 FIRST EVIDENCE GATE

Pair independent physical checks with memory decoding, functional calibration and host-impact measurements.

Layout is schematic, not a construct map. All biological gates remain pending. Published precedents are audited on page 9.

What this revision adds

First research use: classify a defined cultured-cell exposure retrospectively, after its live signal has faded, within a calibrated recording window. The input duration must fit measured write kinetics. No benefit over a simpler recorder has yet been demonstrated.

BLOCK	PROPOSED UPGRADE	EVIDENCE CONTRACT / LIMIT
Maintenance	A separately qualified artificial-chromosome core; candidate layout and copy distribution recorded.	Replication, centromere-mediated segregation, structure and location remain separate endpoints.
One sensor	A defined host-state input, an independent reference and a declared observation window.	Specificity, false activation, sensitivity, latency and perturbation are measured in the chosen host background.
Finite archive	Start with one persistent event bit: unwritten or written. External DNA decoding reads history.	No counts, exact timestamps, universal clock or automatic reset. A written latch is saturated for later events.
Live reporter	A regulated, low-burden indicator of current input-linked state; separate from archive decoding.	Signal intensity does not certify chromosome presence, full integrity or a correctly written archive.
External verifier	An evidence ledger links each assay to its cell, lineage, clone or cohort and carries its uncertainty.	The ledger interprets measurements. It supplies no intracellular governor or molecular authentication mechanism.
Paired system	Requalify function, host burden, physical maintenance and memory after modules are coupled.	Component performance does not transfer automatically to the combined construct.

The added chromosome must earn its complexity: compare useful archival information, host burden, physical stability and the intended extrachromosomal boundary against a simpler reporter/recorder. No requirement says it must win every metric.

The measurements that separate success from appearance

Use independent physical and functional evidence. The table defines comparisons and observables, not molecular construction or laboratory procedures. Acceptance limits require a chosen host, input, window, decision and detection model.

COMPARISON	OBSERVABLES	WHAT IT HELPS DISTINGUISH
Host-only reference	Chosen input and host-state reference; growth, viability and relevant state markers.	Natural response and variation that could be mistaken for an ECHO-M effect.
Chassis / reporter comparison	Physical location, structure, copy distribution, expression and host effects without active recording.	Maintenance or expression burden independent of writing.
Writing-inactive comparison	Input-linked signal and host effects with recording activity absent; verify how well burden is matched.	Sensor cross-talk, writer-associated burden and misleading “matched” controls.
Full system, input absent	Blank/recorded/unknown outcomes; physical retention; unrelated host-state variation.	False writes, spontaneous changes, silencing and reference-reporter common failures.
Full system, input present	Known input history, archive decode, missed events, latency distribution and live-signal trajectory.	Whether the intended event actually reaches the archive in the declared window.
Longitudinal cohort accounting	Enrolled and recovered units, deaths, losses, clone sizes, observation windows and selection status.	Survivor bias and clonal outgrowth that can imitate persistence or repeated writing.

Hard gate before tradeoff scoring: evaluate location / unintended integration and other specified boundary failures with reported detection limits. “No intended host targeting” is an intention, not a measured absence of host effects.

A blank archive is an observation, not a verdict

INTERPRETATION	REQUIRED EVIDENCE	PERMITTED CLAIM
RECORDED	A defined archive mark is decoded, with its error model and physical context documented.	Supports a recorded event within the tested specificity and window; it does not recover later events.
QUALIFIED BLANK	No event mark is recovered and relevant recording competence is supported at the stated observation unit.	Means no recorded event within tested sensitivity and window. It does not prove the input never occurred.
UNKNOWN	Missing / discordant assays, unqualified silencing, loss, incomplete follow-up or insufficient competence evidence.	Supports no event-absence claim. Keep deaths and losses visible in the enrolled-cohort accounting.

ASSAY LINKAGE

Cell / lineage / clone / cohort are different scopes. A destructive assay on one cell cannot certify another live cell. Sister-cell or cohort evidence supports only the explicitly validated inference at that level.

- Store construct revision, host background, input/window, sampling unit, assay provenance, uncertainty, censoring and interpretation.
- External timestamps identify exposures and observations. This ledger is not an intrinsic clock.

KEEP THE WHOLE DENOMINATOR

Report qualified coverage $\mathbf{w = N_{qualified} / N_{enrolled}}$ alongside fidelity. In known-history comparisons, report correct qualified archives per enrolled unit; do not report only the bright survivors.

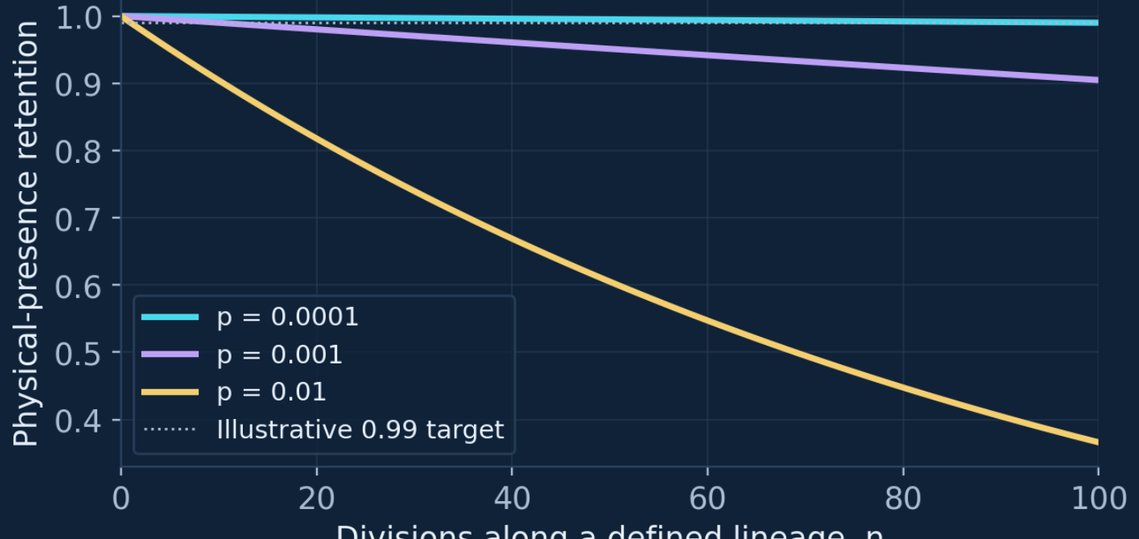
For an event probability q, with unknown outcomes left unconstrained:

$$q_{all} = wq_Q + (1-w)q_{unknown} \;\leq\; wq_{Q,U} + (1-w)$$

Illustration: if 900 of 1,000 units qualify, zero detections in the 900 with known s = 0.8 gives $q_{Q,U}$ = 0.004154 at 95%. The worst-case all-cohort upper bound is **0.10374**. Unknown outcomes dominate. These are calculated examples, not ECHO-M results.

The bound assumes fixed, correctly known coverage and a valid conditional event bound; uncertainty in coverage and qualification also requires statistical treatment.

Retention: specify what survives and who is counted



Illustrative constant-hazard curves; not measured ECHO-M retention.

PHYSICAL PRESENCE ALONG A LINEAGE

$$R_{phys}(n) = \prod_{j=1}^{n}(1 - h_j)$$

h_j is loss probability at division j, conditional on retention through j-1 and the declared history. Constant conditional hazard p gives $(1\text{-}p)^n$. This identity does not require independent successive divisions.

Illustrative target only: R(100) ≥ 0.99 implies $p \leq 1.00498 \times 10^{-4}$ per division for that one endpoint and model. A design target is not a measured loss rate or a safety limit.

Track copy-number distribution, integrity, location and functional expression separately. If reacquisition, changing copy states or heterogeneity matters, use an appropriate multistate model.

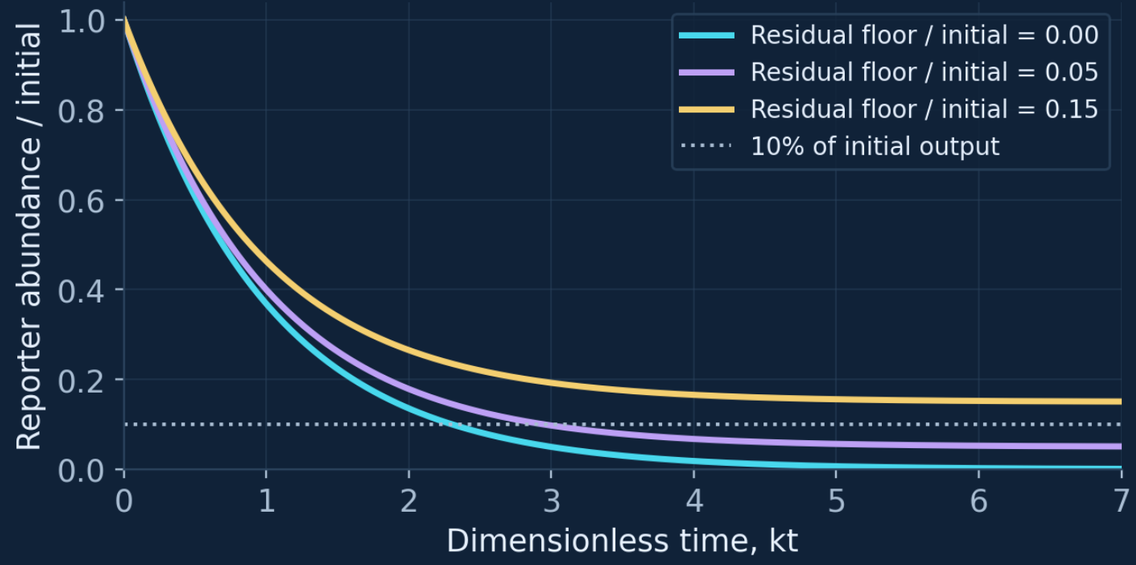
JOINT USEFUL HISTORY IS NOT A PRODUCT OF MARGINAL SCORES

$$P(A \cap I \cap F \cap D) = P(A)P(I \mid A)P(F \mid A, I)P(D \mid A, I, F)$$

A = DNA present; I = required integrity and location; F = qualified function; D = correctly decoded history. Every conditioning term matters. Host burden remains separately reported. Orthogonal measurements can still share host-driven failures.

A proposed population-bias model is $\mathbf{dx/dt = -\lambda_{loss}x + (r_+ - r_-)x(1-x)}$, with all rates per time and x the carrier fraction in a declared population. Fitness differences can mask or exaggerate loss; mortality and missing outcomes need explicit accounting.

Shutoff: a residual floor changes the answer



Dimensionless illustrative model; no molecular time constant is claimed.

FIRST-ORDER MODEL, IF DATA SUPPORT IT

$$\frac{dC}{dt} = b - kC, \qquad C_\infty = \frac{b}{k}$$

$$C(t) = C_\infty + (C_0 - C_\infty)e^{-kt}$$

Define the reporter quantity being measured. k = δ + g models degradation plus growth dilution only under the stated constant-rate assumptions. $g = \ln(2)/T_d$ requires balanced exponential growth; arrested cells need a different fitted model.

ln(10)/k removes 90% of the **excess above the floor**. It removes 90% of total initial output only when b = 0. Estimate the floor independently. Carryover transcripts or reporter maturation may require a multicomponent model.

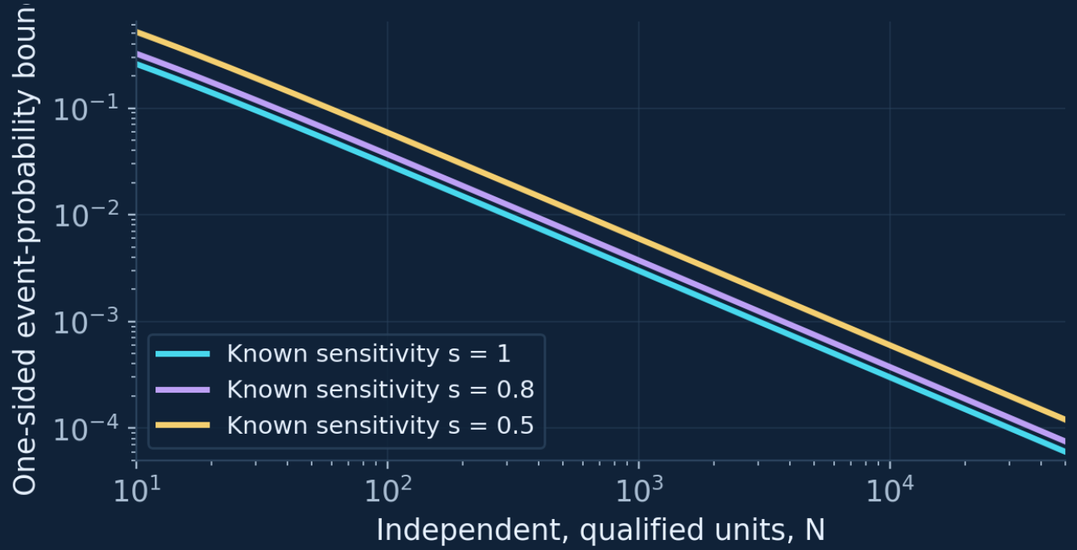
TIME TO A TOTAL-OUTPUT TARGET: C(T) ≤ θC₀

$$T = \frac{1}{k}\ln\left(\frac{C_0 - C_\infty}{\theta C_0 - C_\infty}\right), \quad C_\infty < \theta C_0 < C_0$$

For a 10% target: zero floor gives T = 2.303/k; a 5% floor gives T = 2.944/k; a floor at or above 10% gives **no finite crossing**. A falling mean also does not bound the fraction of cells with residual activity.

Keep four endpoints separate: production OFF; residual output below a defined threshold; no detectable ECHO-M DNA at specified sensitivity; host recovery. Reporter fading cannot establish chromosome clearance or host recovery.

Zero detections: confidence belongs to a defined population



Fixed sensitivity and independent units; curves are analytical examples.

N	KNOWN s	q UPPER	PERCENT
300	1.0	0.009936	0.994%
300	0.8	0.012420	1.242%
1,000	1.0	0.002991	0.299%
1,000	0.8	0.003739	0.374%

To bound q ≤ 0.001 with zero detections: at least 2,995 independent units for s = 1; 3,744 for s = 0.8. Fixed-model sizing only.

KNOWN-SENSITIVITY ZERO-EVENT BOUND

$$P(0\ detected) = (1 - sq)^N$$

$$q_U = \min\left\{1, \frac{1 - \alpha^{1/N}}{s}\right\}$$

For one-sided 95%, α = 0.05. Assumptions: a predeclared endpoint/exposure, independent units with common event probability q, known event-conditioned sensitivity s and no false positives. These are per-unit probabilities over that window, not per-division rates.

If sensitivity varies with the same host state that causes loss or failure, calibrate those strata and their sampling. A nominal sensitivity from easy positive controls may not apply to the missed lineages.

UNCERTAIN SENSITIVITY / RELATED LINEAGES

- Do not stack a 95% sensitivity interval and a 95% event bound and call the result jointly 95%. Allocate error ($\alpha_{event} + \alpha_s \leq 0.05$), or use a justified joint analysis.
- A design-effect effective sample size is not an exact substitute in a zero-count binomial bound. Use independent units or a justified cluster / hierarchical model.
- Report conditional fidelity, qualification coverage, deaths and censored outcomes. A conditional bound is not an all-cohort or universal safety claim.

Court disposition and the next useful proof

REVIEWER	USEFUL CHALLENGE	REVISION DISPOSITION
Grok	Shared host-driven failure and a simpler-recorder comparator.	Adopted: physical checks and a benefit comparison. A second functional reporter was rejected as integrity proof.
Claude	Units, conditional survival, residual floors, missing outcomes and confidence coverage.	Corrected per-division / per-time mixing and conditioning. An effective-N shortcut was withdrawn as an exact tool.
Gemini	Component precedents and the gap between maintenance and recording.	Discarded arbitrary numeric targets and guaranteed-insulation / total-integrity reporter claims. The erratum is flagged.
Meta AI	Host burden, clonal outgrowth, assay linkage and unknown histories.	Adopted: explicit observation scope and full-cohort accounting. Added host-genome barcodes are not required.

Preserved disagreement: a one-event latch deliberately stops adding historical information after its first write. The ledger supplies no clock. Shared physical fate remains a real limitation; the benefit of a HAC over simpler recorders is still unproved.

ALL BIOLOGICAL GATES REMAIN PENDING

1 DEFINE

Choose one host background, input, window and decision. Specify margins, detection limits and sampling units.

2 QUALIFY

Establish physical chassis behavior, low-burden sensing, finite recording and live output as distinct observables.

3 COUPLE

Recheck the assembled system, shared failures, unknown outcomes, output decay and DNA persistence.

4 COMPARE

Demonstrate useful archived history versus a simpler recorder before expanding inputs, event counts or resolution.

A decisive negative result: writing or maintenance removes the useful history advantage, or host burden exceeds the prespecified research margin.

Evidence audit, provenance and credits

[1] Gambogi et al. 2024 - single-copy HAC precedent

Science 383, 1344-1349. doi:10.1126/science.adj3566

Peer-reviewed chassis precedent in studied cell contexts. It does not establish ECHO-M or transfer its performance to an active recorder.

[2] Science erratum, 12 March 2026

Science 391, eaeh0738. doi:10.1126/science.aeh0738

Existence and bibliographic linkage verified in PubMed. Full correction text was inaccessible in this audit; no quantitative formation-efficiency claim is imported.

[3] Birchak et al. 2025 - modular HAC constructs

Chromosome Research 33, 29. doi:10.1007/s10577-025-09788-w

Peer-reviewed precedent for modular construct assembly, sequence validation and functional HAC formation. It is not validation of coupled sensing, memory and reporter behavior.

[4] Choi et al. 2022 - DNA Typewriter

Nature 608, 98-107. doi:10.1038/s41586-022-04922-8

Peer-reviewed sequential DNA-recording precedent. Its discussion places prime-editing recording on days-scale kinetics; ECHO-M rapid capture and a chromosome-contained recorder are not established.

[5] Chen et al. 2024 - ENGRAM

Symbolic recording of signalling and cis-regulatory element activity to DNA. doi:10.1038/s41586-024-07706-4

Peer-reviewed biological-signal-to-DNA recording precedent. Its genomic recording approach supports the conceptual bridge, while self-recording on a maintained ECHO-M chromosome remains untested.

[6] Birchak et al. 2026 - protein-only centromere seeding

bioRxiv preprint, 29 January 2026. doi:10.64898/2026.01.28.702397

Retrieved as a preprint. An optional research direction, not a peer-reviewed prerequisite or an ECHO-M validation claim. No performance advantage is assumed here.

Credits: Anubis - concept and design direction. Asteris Solenne - synthesis, source audit, mathematical corrections and revised brief. Grok, Claude, Gemini and Meta AI - two rounds of adversarial AI-assisted review through the supplied Court tabs. Their outputs were checked and selectively adopted; this is not independent experimental validation or institutional peer review.

Continuity: preserve host dependence, no intended host-genome editing, finite self-recording, reporter-only cultured-cell scope and separate replication / segregation. Existing C_{phys}, C_{expr}, B_{host}, $S_{\times}$ and O objective blocks remain distinct; integration surveillance is a gate, not a tradeoff score. No RNA/chassis optimization fronts are merged.